

Table 3-3 Reduced-Scope Projected Costs by Product Development Area, Phase 1: Development of Performance Assessment Guidelines

Product Development Area/Cost Element	Cost (\$1,000)
Structural Performance Products	\$785
Nonstructural Performance Products	786
Risk Management Products	965
Project Management, Administration, Oversight (Review), and Other Costs (Travel, Communications, Supplies and Equipment)	2462
Project Total	\$4998

Table 3-4 Reduced-Scope Projected Program Costs by Task and Year (\$1,000); Phase 1

Task	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Total
SPP-1 Structural Input to Performance Assessment Guidelines	\$8	-	\$33	-	\$42	\$42	\$33	158
SPP-2 Identify Structural Engr. Demand Parameters)	-	20	20	-	-	-	-	40
SPP-3 Identify Intensity Measures	-	20	10	-	-	-	-	30
SPP-4 Prepare Analysis Guidelines	-	10	30	75	-	-	-	115
SPP-5 Identify Structure Damage Measures	70	70	20	-	-	-	-	160
SPP-6 Structural Input to Model Building Studies	17	33	-	17	33	33	-	133
SPP-7 Dev. Procedures for Input to Nonstructural Evaluation	-	-	-	50	-	-	-	50
SPP-8 Develop Structural Loss Functions	-	33	33	33	-	-	-	99
SPP Total	95	186	146	175	75	75	33	785
NPP-1 Nonstr. Input to Performance Assessment Guidelines	8	-	33	-	42	42	33	158
NPP-2 Develop Catalog of Nonstructural Components	20	-	-	-	-	-	-	20
NPP-3 Identify Nonstructural Performance Measures	-	65	-	-	-	-	-	65
NPP-4 Identify Input EDPs for Nonstructural Components	15	30	-	-	-	-	-	45
NPP-5 Develop Nonstructural Performance Database	-	-	50	40	-	-	-	90
NPP-6 Dev. Nonstructural Performance Evaluation Protocols	20	-	-	-	-	-	-	20
NPP-7 Simplify Nonstructural EDPs	10	10	-	-	-	-	-	20
NPP-8 Develop Procedures for Computing Nonstr. Damage	-	-	65	45	85	-	-	195
NPP-9 Develop Nonstructural Loss Functions	-	-	10	10	10	10	-	40
NPP-10 Nonstructural Input to Model Building Studies	17	33	-	17	33	33	-	133
NPP Total	90	138	158	112	170	85	33	786
RMP-1 Develop Input to Performance Assessment Guidelines	8	-	33	-	42	42	33	158
RMP-2 Develop Structural Loss Functions	5	10	-	-	-	-	-	15
RMP-3 Develop Nonstructural Loss Functions	10	20	55	45	45	40	-	215
RMP-4 Develop Model for Aggregating Losses	-	-	-	17	33	33	-	83
RMP-5 Formulate Concept Aggregation Procedures	17	33	-	-	-	-	-	50
RMP-6 Proc. for Aggregating Local Effects to Global	-	39	40	40	-	-	-	119
RMP-7 Develop Loss Integration Procedures	-	-	20	75	75	100	-	270
RMP-8 Identify Stakeholder Needs	-	15	20	-	-	-	-	35
RMP-9 Develop Standard Performance Characterizations	-	10	10	-	-	-	-	20
RMP Total	40	127	178	177	195	215	33	965
Project Management, Oversight & Administration	255	240	230	230	230	240	120	1545
Other Direct Costs	120	58	137	157	180	135	130	917
Annual Totals	\$600	\$749	\$849	\$851	\$850	\$750	\$349	\$4998

Phase 1: Developing a Methodology for Assessing the Seismic Performance of Buildings—Technical Tasks

4.1 Phase 1 Structural Performance Products Tasks

The purpose of the Structural Performance Products tasks is to develop a practical and reliable generalized methodology for assessing the performance of building structural systems, including: identification of appropriate intensity measures for use in analysis; identification of preferred approaches to employing structural analysis to predict building response as a function of intensity; and methods of predicting structural damage as a function of building response. The Structural Performance Products tasks will be performed in coordination with the Risk Management Product tasks related to calculating losses as a function of structural and nonstructural damage, and the Nonstructural Performance Products tasks that predict nonstructural performance as a function of both ground shaking intensity and structural response. Finally, the generalized methodology will be developed into a straightforward and practical series of implementation guidelines. The first guidelines to be developed will be those for which adequate research results are currently available. For the purposes of this Program Plan, four representative systems are planned to be used. Systems for which there would appear to be adequate research data available and which, in general, cover the range of structural systems and behaviors are:

- Steel moment frames
- Concrete moment frames
- Cantilever concrete wall systems
- Wood wall systems

Should adequate research data and funding be available, this effort could be extended to include other systems, including braced steel frames, coupled concrete walls, and various forms of masonry wall systems.

The Structural Performance Products Team consisting of a team leader, a researcher knowledgeable in the state of art and state of practice of structural analysis, and at least one engineer or researcher with extensive expertise and knowledge of the behavior and design requirements for each of the structural systems will perform these tasks. The team will include at least one expert practicing structural engineer routinely engaged in the design, evaluation, and upgrade of buildings. The Structural Performance Products Team will retain graduate student assistants and consultants to perform many of the development tasks.

Figure 4-1 presents the Structural Performance Products tasks that will be undertaken in Phase 1, along with a schedule for their completion (see also Table 3-1). These SPP tasks are detailed in the sections below.

Phase 1: Task Description		Year 1	Year 2	Year 3	Year 4	Year 5
SPP-1	Structural Input to Engineering Guidelines					
SPP-2	Identify Structural Engineering Demand Parameters					
SPP-3	Identify Intensity Measures					
SPP-4	Prepare Analysis Guidelines					
SPP-5	Identify Structural Damage Measures					
SPP-6	Structural Input to Model Building Studies					
SPP-7	Develop Procedures for Input to Nonstructural Evaluation					
SPP-8	Develop Structural Loss Functions					

Figure 4-1 Phase 1: Schedule for structural performance products tasks.

4.1.1 *Structural Input to Performance Assessment Guidelines (SPP-1)*

The development of performance assessment guidelines will be jointly performed by the three Product Development Teams. The Structural Performance Products Team will be responsible for developing those portions of the guidelines relating to characterizing ground shaking hazards through intensity measures, modeling and analyzing structures to derive building-specific structural response functions and to develop input data for the evaluation of nonstructural components and systems, identification of structural damage measures, and development of structural fragilities.

The performance assessment guidelines will be developed incrementally over the five-year span of Phase 1. Preliminary draft guidelines will be developed in the first year of the program. This will enable the Structural Performance